Suyash Maniyar

(413) 409-9018 | smaniyar@umass.edu | linkedin.com/in/maniyar-suyash | github.com/SuyashManiyar | Portfolio

Education

University of Massachusetts Amherst — Amherst, MA, USA
Master of Science in Computer Science ***(CGPA: 4.0/4.0)*** — *Expected May 2027*

Indian Institute of Technology, Jodhpur — Jodhpur, India
Bachelor of Technology in Electrical Engineering ***(CGPA: 8.2/10.0)*** — *Aug 2019 - May 2023*

Coursework: Data Structures & Algorithms, Advanced NLP, AI Alignment, Systems for Deep Learning, Deep Learning, Machine Learning, DLOps, Computer Networks, Software Engineering, Systems for Data Science

Publications and Preprints

- Measuring Reward Hacking and Reasoning-Answer Decoupling Under Confounded Optimization, **COLM 2026**
- AI-Generated Lecture Slides for Improving Slide Element Detection and Slide Image Retrieval, **ICDAR 2025**
- Enhancing Legal LLMs with Metadata-Enriched Retrieval and DPO, **arXiv preprint**

Professional Experience

Applied AI and Data Science Co-op — July 2026 - Present
Johnson & Johnson MedTech — *Danvers, MA, USA*

- Built a Django **human-in-the-loop annotation platform** that pre-fills LLM extractions with bounding-box evidence, letting annotators approve or edit rather than label from scratch, cutting annotation time by **80%**.
- Developed an **agentic** clinical information extraction pipeline (OCR tool calls, LLM extraction, LLM-as-a-judge) to automate annotation process,with 97% field accuracy, routing only low-confidence cases back to human review.
- Deployed on **Azure** App Service and **PostgreSQL** Flexible Server, with batch OCR in a Container Apps job scaling to 30 replicas at 5 pages/second; CI/CD through GitLab pipelines.
- Designed an **anomaly detection** pipeline on **250 Hz** implanted cardiac sensor time-series data, improving Right Ventricular Dysfunction **F1 by 7%**; developing **cardiac output regression** from sensor-derived features.
- Fine-tuned open-weight LLMs and VLMs (Llama 3.2 3B/11B) for medical NER on scanned clinical PDFs.

Advanced Data Science Associate — Nov 2024 - July 2025
ZS Associates — *Pune, India*

- Developed a **Switch Prediction Model** to predict patients switching from ongoing drug treatments, leveraging clinical data, and an ensemble of ML (Bayes, XGBoost), DL (TabNet) models, improving F1 score by **11%.**
- Engineered an end-to-end **MLOps** pipeline on **AWS** (SageMaker) orchestrated with **Airflow** and **Docker**, automating **drift detection and retraining** across 5+ models, cutting manual deployment effort by **60%**.
- Performed SHAP analysis and aided in optimizing marketing with projected cost savings of up to **500K dollars**.
- Built patient cohorts from **100M+** claims records by applying business rules in **SQL on Amazon Redshift**.
- Quantified impact of advertisements using **A/B testing** and **Double ML**, enabling optimized campaign spend.

ML Engineer — July 2023 - Oct 2024
Decimal Point Analytics — *Mumbai, India*

- Built an XGBoost **supply chain optimization** model for a leading FMCG client, forecasting SKU-level demand from lag and rolling-window features engineered in **Spark**, cutting monthly inventory waste by **15%**.
- Designed an **Anomaly Detection** pipeline to predict defects in industrial machinery using time series data.
- Shipped a **RAG** based **chatbot** for Q&A, leveraging LangChain and Pinecone vector database; integrated knowledge graphs, increasing accuracy by 9%. Implemented FAQ response **caching**, reducing inference cost by 30%.
- Automated an internal process of **web scraping** by fine-tuning Llama 2 (7B/13B) with LoRA for text-based extraction and leveraging GPT-4 for **code generation pipeline**, reducing downtime from **12 hours to 3 mins**.
- Built a **computer vision** based car damage segmentation system, improving claim processing efficiency by 75%.
- Compressed few-shot object detection models by **70%** using **knowledge distillation** enabling edge deployment.

Machine Learning Intern — Aug 2022 - May 2023
RaapidAI — *Remote, India*

- Built a **table detection and structure recognition** pipeline for medical document images, fine-tuning CascadeTabNet, DETR and DiT to **76% mAP**, with augmentation adding **7%** on a private gold-standard set.

Research Experience

Reward Hacking Under Position-Confounded RL : COLM 2026 Workshop Feb 2026 - May 2026

- **Post-trained** 10 LLMs (Qwen2.5, Llama 3.x) via **GRPO** on confounded reward, inducing **reward hacking**.
- Identified a **backdoor-like failure** from fully correct training data that endpoint metrics cannot detect, and designed an **evaluation protocol** including reasoning-answer decoupling, out-of-domain probes to detect it.
- **Red-teamed** checkpoints: the hacked model chose the **harmful option 84%** of the time in the rewarded position vs **4%** for controls, and **0%** once position-swapped.

Is the Refusal Direction Mostly Harmfulness? : TMLR (review) Feb 2026 - Aug 2026

- Reproduced the claim that LLMs encode harmfulness and refusal separately on **8 LLMs** (Llama-3, Qwen2.5, Gemma-3, **3B to 27B**) using **nnsight** activation interventions.
- Showed the separation fails: after removing the refusal part, the harm direction keeps **76% of its causal effect**.
- Broke the proposed **latent safety guard** with a **nanoGCG** suffix (recall down to **40-74%**).

AI Alignment as a Continual Learning Problem Jan 2026 - May 2026

- Studied whether LLMs can both **learn and unlearn** behaviors with **LoRA**, training **6 sequential adapters** on paired **safety, bias, truthfulness and privacy** tasks (PKU-SafeRLHF, BBQ, TruthfulQA, TOFU).
- Found all **4 merge strategies** collapsed refusal to **0.005**; built a **zero-parameter centroid router** that beat the best merge by up to **52 points** and held MMLU at **0.406** vs **0.170**.

Metadata-Enriched Hybrid RAG + DPO for Legal LLMs : arXiv Sept 2025 - Dec 2025

- Built **hybrid FAISS retrieval** fusing dense and **BM25** scores, cutting retrieval mismatch from **65% to 10%**.
- Engineered **metadata-enriched chunking** using case summaries and details, improving **recall@10 by 23%**.
- Aligned **Llama 3.2 1B** via **DPO** on 3.8K preference pairs, cutting **over-refusal** from **53.2% to 1.5%**.

Agentic RL Self-Play for Biomedical QA : BioNLP Lab (Prof. Hong Yu) Aug 2025 - Dec 2025

- Designed an ontology-driven **agentic RL self-play** framework with dual Q&A LLM agents, using **MCP** for tool-grounded verification, improving biomedical QA by **12%** over fine-tuning.

Synthetic Lecture Slides for Detection and Retrieval : ICDAR 2025 Aug 2024 - Feb 2025

- Built an **LLM-guided synthetic data pipeline** (GPT-4, 18 layouts) generating **4,400 auto-labelled slides** across 16 element classes with no manual labelling.
- Fine-tuned **CLIP** on synthetic data to **R@1 43** on real slide retrieval, beating 31k real out-of-domain slides (R@1 31); lifted **LayoutLMv3** detection from 27.3 to **33.9 mAP**.

Projects

EmbedServe : High-Performance Embedding Serving System for Rec Sys

- Built a high-performance embedding serving system in C++, integrating RocksDB, and dynamic request batching.
- Reduced memory by **3.5**× and p99 latency by **3.5**× at ∼2,000 req/s through tiered storage, INT8 quantization.
- Achieved **45% of peak GPU bandwidth** with a fused **Triton kernel**, matching TorchRec within **20% p99 latency** using 1.55× less memory.

UniTrade : Campus Marketplace Web Application : Github link

- Built a campus-only marketplace on a **Next.js/TypeScript** client and **Node/Express REST API**.
- Implemented live bidding over **WebSockets (Socket.IO)**, pushing bid updates to connected clients in real time.
- Added natural-language search using NLP to parse text queries into structured category, price, condition filters.

Slide Scribe : Winner, HackUMass XIII : Project link

- Built real-time **voice Q&A** over live lecture slides for visually impaired students, detecting slide changes via **CLIP** embedding similarity and answering with **Gemini**; Flutter client, FastAPI backend.

Cohort Analysis & Engagement Segmentation : **Education Technology - B.Tech Project**

- Built an **engagement classification** framework on Edpuzzle watch-time and quiz analytics for students enrolled in the Intro to Engineering course, using **k-means** to flag the low-engagement **10%** for instructor intervention.
- Evaluated experiments comparing quiz performance between cohorts with and without video assignments.

Technical Skills

Languages: Python, SQL, C/C++, TypeScript, JavaScript, HTML/CSS, R, LaTeX
LLMs & Alignment: GRPO, DPO, RLHF, SFT, LoRA/QLoRA, Hugging Face (TRL, PEFT), nnsight, RAG
Software Engineering: Django, Node/Express, Next.js, React, REST APIs, FastAPI, Flutter, Distributed Systems
Cloud & Infra: AWS (EC2, SageMaker), GCP, Azure, Docker, CI/CD (GitLab, Jenkins)
ML & AI: PyTorch, TensorFlow, Vector DBs, Kubernetes, Spark, Hadoop, ETL Orchestration, SLURM, CUDA